

CohortCharacteristics

An R package to characterise cohorts

2026-09-17

CohortCharacteristics



Website

CohortCharacteristics is on CRAN:

```
install.packages("CohortCharacteristics")
```

The package documentation and vignettes are available on the website:

<https://darwin-eu.github.io/CohortCharacteristics/>

Set up

```
options("width"=115)
library(duckdb)
library(CDMConnector)
library(PatientProfiles)
library(CohortConstructor)
library(dplyr)
library(here)
library(CohortCharacteristics)

cdm <- omock::mockCdmFromDataset(datasetName = "GiBleed",
                                source = "duckdb")
```

Let's get started

Let's create some cohorts that we will use during the presentation:

```
codes <- c(list(anemia = 439777,
               rheumatoid_arthritis = 80809,
               osteoarthritis = 80180,
               pneumonia = 255848))

cdm[["my_cohort"]] <- conceptCohort(cdm = cdm,
                                   conceptSet = codes,
                                   name = "my_cohort")

cdm[["my_cohort"]] |>
  glimpse()
```

Rows: ??

Columns: 4

```
$ cohort_definition_id <int> 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, ...
$ subject_id          <int> 3699, 359, 377, 793, 1950, 2483, 2509, 3582, 3745, 4239, 5317, 250, 1138, 1211, 1776...
$ cohort_start_date   <date> 1999-07-08, 2014-10-28, 2003-02-13, 2010-05-14, 1978-12-17, 1984-08-11, 2006-05-20,...
$ cohort_end_date     <date> 1999-07-08, 2014-10-28, 2003-02-13, 2010-05-14, 1978-12-17, 1984-08-11, 2006-05-20,...
```

Workflow

We have three types of functions:

- **summarise**: these functions produce a standardised summary of a cohort. This output is called `<summarised_result>`.
- **table**: these functions produce tables from the `<summarised_result>` object generated by the *summarise* functions.
- **plot**: these functions produce plots from the `<summarised_result>` object generated by the *summarise* functions. (We won't show these today)

Contents

- summariseCohortCount
- summariseCohortAttrition
- summariseCharacteristics
- summariseLargeScaleCharacteristics
- summariseCohortTiming
- summariseCohortOverlap

summariseCohortCount

Use `summariseCohortCount()` to get counts for your cohort. Pipe your cohort table into the function as shown below.

```
results <- cdm$my_cohort |>
  summariseCohortCount()
```

It returns cohort counts in a `summarised_result` object.

```
glimpse(results)
```

```
Rows: 8
Columns: 13
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed"
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "osteoarthritis", "osteoarthritis", "pneumonia", "pneumonia", "rheum..."
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall"
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall"
$ variable_name  <chr> "Number records", "Number subjects", "Number records", "Number subjects", "Number record..."
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count"
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer"
$ estimate_value <chr> "100", "100", "2630", "2630", "52", "52", "18", "18"
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall"
```

summariseCohortCount

Note

Previously we saw the function `cohortCount()` to see the counts of a `<cohort_table>`.

```
cohortCount(cdm$my_cohort)
```

```
# A tibble: 4 × 3
  cohort_definition_id number_records number_subjects
      <int>          <int>          <int>
1           1           100           100
2           2          2630          2630
3           3           52           52
4           4           18           18
```

The function `summariseCohortCount()` produces a similar result but in a standardised output (`<summarised_result>`).

tableCohortCount

To get a tidy table from the `<summarised_result>` object.

```
tableCohortCount(result = results)
```

CDM name	Variable name	Estimate name	Cohort name			
			anemia	osteoarthritis	pneumonia	rheumatoid_arthritis
GiBleed	Number records	N	100	2,630	52	18
	Number subjects	N	100	2,630	52	18

summariseCohortAttrition

To summarise cohort attrition, pipe your cohort table into the function as shown below.

```
results <- cdm$my_cohort |>
  summariseCohortAttrition()
```

It returns cohort attrition in a **summarised_result** object.

```
glimpse(results)
```

```

Rows: 64
Columns: 13
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"...
$ strata_name    <chr> "reason", "reason", "reason", "reason", "reason", "reason", "reason", "reason", "reason"...
$ strata_level   <chr> "Initial qualifying events", "Initial qualifying events", "Initial qualifying events", "...
$ variable_name  <chr> "number_records", "number_subjects", "excluded_records", "excluded_subjects", "number_re...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ...
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count", "count", "count"...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", ...
$ estimate_value <chr> "102", "102", "0", "0", "100", "100", "2", "2", "100", "100", "0", "0", "100", "100", "0...
$ additional_name <chr> "reason_id", "reason_id", "reason_id", "reason_id", "reason_id", "reason_id", "reason_id"...

```

summariseCohortAttrition



Note

Yesterday we saw the function `attrition()` to see the attrition of a `<cohort_table>`.

```
attrition(cdm$my_cohort)
```

```
# A tibble: 16 × 7
```

	cohort_definition_id	number_records	number_subjects	reason_id	reason	excluded_records	excluded_subjects
	<int>	<int>	<int>	<int>	<chr>	<int>	<int>
1	1	102	102	1	Initial qualif...	0	0
2	1	100	100	2	Record in obse...	2	2
3	1	100	100	3	Not missing re...	0	0
4	1	100	100	4	Merge overlapp...	0	0
5	2	2694	2694	1	Initial qualif...	0	0
6	2	2630	2630	2	Record in obse...	64	64
7	2	2630	2630	3	Not missing re...	0	0
8	2	2630	2630	4	Merge overlapp...	0	0
9	3	52	52	1	Initial qualif...	0	0
10	3	52	52	2	Record in obse...	0	0
11	3	52	52	3	Not missing re...	0	0
12	3	52	52	4	Merge overlapp...	0	0
13	4	18	18	1	Initial qualif...	0	0
14	4	18	18	2	Record in obse...	0	0
15	4	18	18	3	Not missing re...	0	0
16	4	18	18	4	Merge overlapp...	0	0

The function `summariseCohortAttrition()` produces a similar result but in a standardised output (`<summarised_result>`).

tableCohortAttrition

To get a tidy table from the `<summarised_result>` object.

```
tableCohortAttrition(result = results)
```

Reason	Variable name			
	number_records	number_subjects	excluded_records	excluded_subjects
GiBleed; anemia				
Initial qualifying events	102	102	0	0
Record in observation	100	100	2	2
Not missing record date	100	100	0	0
Merge overlapping records	100	100	0	0
GiBleed; osteoarthritis				
Initial qualifying events	2,694	2,694	0	0
Record in observation	2,630	2,630	64	64
Not missing record date	2,630	2,630	0	0
Merge overlapping records	2,630	2,630	0	0
GiBleed; pneumonia				
Initial qualifying events	52	52	0	0
Record in observation	52	52	0	0
Not missing record date	52	52	0	0
Merge overlapping records	52	52	0	0
GiBleed; rheumatoid_arthritis				
Initial qualifying events	18	18	0	0
Record in observation	18	18	0	0

summariseCharacteristics

The main functionality of the package is to extract cohort characteristics with `summariseCharacteristics()`.

```
results <- cdm$my_cohort |>
  summariseCharacteristics()
```

It returns cohort characteristics in a `<summarised_result>` object.

```
glimpse(results)
```

Rows: 204

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name      <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name    <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level   <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"...
$ strata_name   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level  <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name <chr> "Number records", "Number subjects", "Cohort start date", "Cohort start date", "Cohort s...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, "Female", "F...
$ estimate_name <chr> "count", "count", "min", "q25", "median", "q75", "max", "min", "q25", "median", "q75", "...
$ estimate_type <chr> "integer", "integer", "date", "date", "date", "date", "date", "date", "date", "date", "date", "d...
$ estimate_value <chr> "100", "100", "1912-04-30", "1974-01-30", "1990-12-04", "2005-08-09", "2018-11-08", "191...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```

tableCharacteristics

To get a tidy table from the `<summarised_result>` object.

```
tableCharacteristics(result = results, header = c("cdm_name", "cohort_name"))
```

			CDM name			
			GiBleed			
			Cohort name			
Variable name	Variable level	Estimate name	anemia	osteoarthritis	pneumonia	rheumatoid_arthritis
Number records	–	N	100	2,630	52	18
Number subjects	–	N	100	2,630	52	18
Cohort start date	–	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2000-08-10 [1992-01-18 - 2008-01-01]
		Range	1912-04-30 to 2018-11-08	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
Cohort end date	–	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2004-11-17 [1993-03-05 - 2011-05-22]
		Range	1912-04-30 to 2018-11-08	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
Age	–	Median [Q25 - Q75]	32.00 [17.75 - 50.25]	38.00 [36.00 - 41.00]	82.00 [78.00 - 86.25]	41.00 [34.25 - 55.00]
		Mean (SD)	33.49 (22.14)	38.15 (3.27)	80.33 (9.48)	45.22 (15.11)
		Range	0.00 to 81.00	31.00 to 46.00	53.00 to 96.00	24.00 to 77.00
Sex	Female	N (%)	78 (78.00%)	1,341 (50.99%)	36 (69.23%)	8 (44.44%)
	Male	N (%)	22 (22.00%)	1,289 (49.01%)	16 (30.77%)	10 (55.56%)
Prior observation	–	Median [Q25 - Q75]	12,045.00 [6,843.00 - 18,706.00]	14,097.00 [13,161.25 - 15,039.00]	30,079.00 [28,538.25 - 31,726.50]	15,330.00 [12,866.25 - 20,440.00]
		Mean (SD)	12,570.42 (8,107.27)	14,105.97 (1,193.14)	29,498.13 (3,467.33)	16,870.94 (5,513.54)
		Range	0.00 to 29,930.00	11,340.00 to 17,044.00	19,485.00 to 35,210.00	9,125.00 to 28,469.00
Future observation	–	Median [Q25 - Q75]	9,641.50 [4,596.00 - 14,747.00]	6,790.00 [3,568.00 - 10,012.00]	12,000 [10,000 - 15,000]	6,065.50 [4,058.00 - 9,716.00]

Commonly used arguments for summariseCharacteristics

- **strata**: creates stratified results.
- **cohortId**: filters by cohort ID.
- **ageGroup**: defines the age groups to summarise.
- **tableIntersect**: count/flag/date/days
- **cohortIntersect**: count/flag/date/days
- **conceptIntersect**: count/flag/date/days
- **link to documentation**

ageGroup example

```
result <- summariseCharacteristics(cdm$my_cohort, ageGroup = list(c(0,10), c(11,18), c(19,150)))  
tableCharacteristics(result = result, header = "cohort_name")
```

ageGroup example

CDM name	Variable name	Variable level	Estimate name	Cohort name			
				anemia	osteoarthritis	pneumonia	rheumatoid_arthritis
GiBleed	Number records	—	N	100	2,630	52	18
	Number subjects	—	N	100	2,630	52	18
	Cohort start date	—	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2000-08-10 [1992-01-18 - 2008-01-01]
			Range	1912-04-30 to 2018-11-08	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Cohort end date	—	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2004-11-17 [1993-03-05 - 2011-05-22]
			Range	1912-04-30 to 2018-11-08	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Age	—	Median [Q25 - Q75]	32.00 [17.75 - 50.25]	38.00 [36.00 - 41.00]	82.00 [78.00 - 86.25]	41.00 [34.25 - 55.00]
			Mean (SD)	33.49 (22.14)	38.15 (3.27)	80.33 (9.48)	45.22 (15.11)
			Range	0.00 to 81.00	31.00 to 46.00	53.00 to 96.00	24.00 to 77.00
	Age group	0 to 10	N (%)	20 (20.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
		11 to 18	N (%)	7 (7.00%)	0 (0.00%)	0 (0.00%)	0 (0.00%)
		19 to 150	N (%)	73 (73.00%)	2,630 (100.00%)	52 (100.00%)	18 (100.00%)
	Sex	Female	N (%)	78 (78.00%)	1,341 (50.99%)	36 (69.23%)	8 (44.44%)
		Male	N (%)	22 (22.00%)	1,289 (49.01%)	16 (30.77%)	10 (55.56%)
	Prior observation	—	Median [Q25 - Q75]	12,045.00 [6,843.00 - 18,706.00]	14,097.00 [13,161.25 - 15,039.00]	30,079.00 [28,538.25 - 31,726.50]	15,330.00 [12,866.25 - 20,440.00]
			Mean (SD)	12,570.42 (8,107.27)	14,105.97 (1,193.14)	29,498.13 (3,467.33)	16,870.94 (5,513.54)
			Range	0.00 to 29,930.00	11,340.00 to 17,044.00	19,485.00 to 35,210.00	9,125.00 to 28,469.00
	Future observation		Median [Q25 - Q75]	9,641.50 [4,596.00 -	6,790.00 [3,568.00 -	12,000.00 [10,000 - 15,000]	6,065.50 [4,058.00 -

Strata example

```
result <- cdm$my_cohort |>
  addAge(ageGroup = list(c(0,10),c(11,18), c(19,150))) |>
  summariseCharacteristics(cohortId = "anemia",
                           strata = list("age_group"))
tableCharacteristics(result = result, header = c("cdm_name", "cohort_name","age_group"))
```

Strata example

			CDM name			
			GiBleed			
			Cohort name			
			anemia			
Variable name	Variable level	Estimate name	Age group			
			overall	0 to 10	11 to 18	19 to 150
Number records	–	N	100	20	7	73
Number subjects	–	N	100	20	7	73
Cohort start date	–	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1963-03-26 [1956-03-03 - 1972-12-27]	1969-08-19 [1963-03-03 - 1975-04-11]	1999-03-30 [1987-11-03 - 2007-06-08]
		Range	1912-04-30 to 2018-11-08	1912-04-30 to 1987-01-08	1951-04-05 to 1986-04-07	1939-01-29 to 2018-11-08
Cohort end date	–	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1963-03-26 [1956-03-03 - 1972-12-27]	1969-08-19 [1963-03-03 - 1975-04-11]	1999-03-30 [1987-11-03 - 2007-06-08]
		Range	1912-04-30 to 2018-11-08	1912-04-30 to 1987-01-08	1951-04-05 to 1986-04-07	1939-01-29 to 2018-11-08
Age	–	Median [Q25 - Q75]	32.00 [17.75 - 50.25]	2.50 [0.75 - 5.25]	17.00 [14.50 - 17.50]	43.00 [28.00 - 56.00]
		Mean (SD)	33.49 (22.14)	3.30 (3.15)	15.86 (2.27)	43.45 (16.93)
		Range	0.00 to 81.00	0.00 to 9.00	12.00 to 18.00	19.00 to 81.00
Sex	Female	N (%)	78 (78.00%)	10 (50.00%)	6 (85.71%)	62 (84.93%)
	Male	N (%)	22 (22.00%)	10 (50.00%)	1 (14.29%)	11 (15.07%)
Prior observation	–	Median [Q25 - Q75]	12,045.00 [6,843.00 - 18,706.00]	1,094.50 [365.00 - 2,280.25]	6,570.00 [5,657.00 - 6,752.00]	16,060.00 [10,585.00 - 20,804.00]
		Mean (SD)	12,570.42 (8,107.27)	1,478.00 (1,221.91)	6,152.57 (827.72)	16,224.85 (6,180.75)
		Range	0.00 to 29,930.00	0.00 to 3,650.00	4,745.00 to 6,935.00	7,300.00 to 29,930.00

Intersection with another table

You can get count/flag/date/days summaries with other tables.

```
result <- cdm$my_cohort |>
  summariseCharacteristics(
    cohortId = "anemia",
    tableIntersectCount = list(
      "Number of visits prior year" = list(
        tableName = "visit_occurrence",
        window = c(-365, 0)
      )
    ),
    cohortIntersectFlag = list(
      "Previous rheumatoid arthritis diagnosis" = list(
        targetCohortTable = "my_cohort",
        targetCohortId = "rheumatoid_arthritis",
        window = c(-Inf, 0)
      )
    ),
    conceptIntersectDays = list(
      "Time since last X-ray" = list(
```

Intersection with another table

You can get count/flag/date/days summaries with other tables.

CDM name	Variable name	Variable level	Estimate name	Cohort name			
				anemia	osteoarthritis	pneumonia	rheumatoid_arthritis
GiBleed	Number records	—	N	100	2,630	52	18
	Number subjects	—	N	100	2,630	52	18
	Cohort start date	—	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2000-08-10 [1992-01-18 - 2008-01-01]
			Range	1912-04-30 to 2018-11-08	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Cohort end date	—	Median [Q25 - Q75]	1990-12-04 [1974-01-30 - 2005-08-09]	1999-11-16 [1988-03-03 - 2008-08-07]	2007-07-06 [1998-09-17 - 2013-04-07]	2004-11-17 [1993-03-05 - 2011-05-22]
			Range	1912-04-30 to 2018-11-08	1941-12-05 to 2019-04-14	1980-03-03 to 2019-02-18	1971-07-29 to 2017-06-06
	Age	—	Median [Q25 - Q75]	32.00 [17.75 - 50.25]	38.00 [36.00 - 41.00]	82.00 [78.00 - 86.25]	41.00 [34.25 - 55.00]
			Mean (SD)	33.49 (22.14)	38.15 (3.27)	80.33 (9.48)	45.22 (15.11)
			Range	0.00 to 81.00	31.00 to 46.00	53.00 to 96.00	24.00 to 77.00
	Sex	Female	N (%)	78 (78.00%)	1,341 (50.99%)	36 (69.23%)	8 (44.44%)
		Male	N (%)	22 (22.00%)	1,289 (49.01%)	16 (30.77%)	10 (55.56%)
	Prior observation	—	Median [Q25 - Q75]	12,045.00 [6,843.00 - 18,706.00]	14,097.00 [13,161.25 - 15,039.00]	30,079.00 [28,538.25 - 31,726.50]	15,330.00 [12,866.25 - 20,440.00]
			Mean (SD)	12,570.42 (8,107.27)	14,105.97 (1,193.14)	29,498.13 (3,467.33)	16,870.94 (5,513.54)
			Range	0.00 to 29,930.00	11,340.00 to 17,044.00	19,485.00 to 35,210.00	9,125.00 to 28,469.00
	Future observation	—	Median [Q25 - Q75]	9,641.50 [4,596.00 - 15,796.25]	6,790.00 [3,568.00 - 10,781.25]	12.00 [10.00 - 15.00]	6,065.50 [4,058.00 - 9,716.00]
			Mean (SD)	10,773.47 (7,880.91)	7,686.76 (5,364.39)	12.25 (3.16)	6,978.83 (4,735.79)
			Range	1.00 to 39,143.00	10.00 to 28,328.00	7.00 to 19.00	668.00 to 17,356.00

Intersections

Note

The arguments of the intersection functions match the *PatientProfiles* functions that we have just seen:

- `tableIntersectFlag -> PatientProfiles::addTableIntersectFlag()`
- `tableIntersectCount -> PatientProfiles::addTableIntersectCount()`
- `tableIntersectDays -> PatientProfiles::addTableIntersectDays()`
- `tableIntersectDate -> PatientProfiles::addTableIntersectDate()`
- `conceptIntersectFlag -> PatientProfiles::addConceptIntersectFlag()`
- `conceptIntersectCount -> PatientProfiles::addConceptIntersectCount()`
- `conceptIntersectDays -> PatientProfiles::addConceptIntersectDays()`
- `conceptIntersectDate -> PatientProfiles::addConceptIntersectDate()`
- `cohortIntersectFlag -> PatientProfiles::addCohortIntersectFlag()`
- `cohortIntersectCount -> PatientProfiles::addCohortIntersectCount()`
- `cohortIntersectDays -> PatientProfiles::addCohortIntersectDays()`
- `cohortIntersectDate -> PatientProfiles::addCohortIntersectDate()`



Your turn!

Exercise 1 - characterise an osteoarthritis cohort

```
cdm <- omock::mockCdmFromDataset(datasetName = "GiBleed",  
                                  source = "duckdb")  
  
cdm[["my_cohort"]] <- conceptCohort(cdm = cdm,  
                                   conceptSet = codes,  
                                   name = "my_cohort")
```

Can you characterise the **osteoarthritis** cohort?

- Start with only demographics
- Add if they had any anemia diagnosis during the last 180 days (including index date).
- Add if they had any pneumonia diagnosis last 2 years (excluding index date).

►  [Click to see solution](#)

summariseLargeScaleCharacteristics

Sometimes we might want to summarise all clinical events for cohorts across different time windows. We can do this with `summariseLargeScaleCharacteristics()`.

```
result <- cdm$my_cohort |>
  summariseLargeScaleCharacteristics(
    window = list(c(-Inf, -1), c(1, Inf)),
    eventInWindow = "condition_occurrence",
    minimumFrequency = 0.05
  )

glimpse(result)
```

Rows : 250

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name"...
$ group_level    <chr> "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia", "anemia"...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "Acute allergic reaction", "Acute allergic reaction", "Acute bronchitis", "Acute bronchi...
$ variable_level <chr> "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-inf to -1", "-in...
$ estimate_name  <chr> "count", "percentage", "count", "percentage", "count", "percentage", "count", "percentag...
$ estimate_type  <chr> "integer", "percentage", "integer", "percentage", "integer", "percentage", "integer", "p...
$ estimate_value <chr> "5", "5.10", "77", "78.57", "75", "76.53", "7", "7.14", "6", "6.12", "6", "6.12", "7", "...
$ additional_name <chr> "concept_id", "concept_id", "concept_id", "concept_id", "concept_id", "concept_id", "con..."
```

tableTopLargeScaleCharacteristics

`tableTopLargeScaleCharacteristics()` creates a tidy table of the most frequent concepts.

```
tableTopLargeScaleCharacteristics(result = result, topConcepts = 5)
```

Top	Cohort name						
	anemia		osteoarthritis		pneumonia	rheumatoid_arthritis	
			Window				
	-inf to -1	1 to inf	-inf to -1	1 to inf	-inf to -1	-inf to -1	1 to inf
1	Viral sinusitis (40481087) 82 (83.7%)	Viral sinusitis (40481087) 82 (82.0%)	Viral sinusitis (40481087) 2594 (98.6%)	Viral sinusitis (40481087) 2066 (78.6%)	Alzheimer's disease (378419) 43 (82.7%)	Acute viral pharyngitis (4112343) 17 (94.4%)	Viral sinusitis (40481087) 13 (72.2%)
2	Acute bronchitis (260139) 77 (78.6%)	Acute viral pharyngitis (4112343) 70 (70.0%)	Acute viral pharyngitis (4112343) 2403 (91.4%)	Acute viral pharyngitis (4112343) 1664 (63.3%)	Streptococcal sore throat (28060) 43 (82.7%)	Viral sinusitis (40481087) 17 (94.4%)	Acute viral pharyngitis (4112343) 12 (66.7%)
3	Acute viral pharyngitis (4112343) 75 (76.5%)	Acute bronchitis (260139) 65 (65.0%)	Acute bronchitis (260139) 2292 (87.2%)	Acute bronchitis (260139) 1515 (57.6%)	Otitis media (372328) 41 (78.8%)	Otitis media (372328) 13 (72.2%)	Acute cholecystitis (198809) 1 (5.6%)
4	Otitis media (372328) 69 (70.4%)	Osteoarthritis (80180) 58 (58.0%)	Otitis media (372328) 1935 (73.6%)	Chronic sinusitis (257012) 251 (9.5%)	Sprain of ankle (81151) 35 (67.3%)	Osteoarthritis (80180) 12 (66.7%)	Alzheimer's disease (378419) 1 (5.6%)
5	Streptococcal sore throat (28060) 59 (60.2%)	Angiodysplasia of stomach (4310024) 9 (9.0%)	Streptococcal sore throat (28060) 1399 (53.2%)	Cerebrovascular accident (381316) 194 (7.4%)	Laceration of foot (4109685) 5 (9.6%)	Streptococcal sore throat (28060) 9 (50.0%)	Anemia (439777) 1 (5.6%)

tidy

You can create a tidy table that you can explore with the `tidy()` function:

```
tidy(result)
```

```
# A tibble: 125 × 10
```

	cdm_name	cohort_name	variable_name	variable_level	concept_id	count	percentage	analysis	table_name	type
	<chr>	<chr>	<chr>	<chr>	<chr>	<int>	<dbl>	<chr>	<chr>	<chr>
1	GiBleed	anemia	Acute allergic reacti...	-inf to -1	4084167	5	5.1	standard	condition...	event
2	GiBleed	anemia	Acute bronchitis	-inf to -1	260139	77	78.6	standard	condition...	event
3	GiBleed	anemia	Acute viral pharyngit...	-inf to -1	4112343	75	76.5	standard	condition...	event
4	GiBleed	anemia	Angiodysplasia of sto...	-inf to -1	4310024	7	7.14	standard	condition...	event
5	GiBleed	anemia	Appendicitis	-inf to -1	440448	6	6.12	standard	condition...	event
6	GiBleed	anemia	Child attention defici...	-inf to -1	440086	6	6.12	standard	condition...	event
7	GiBleed	anemia	Concussion with loss ...	-inf to -1	375671	7	7.14	standard	condition...	event
8	GiBleed	anemia	Cystitis	-inf to -1	195588	7	7.14	standard	condition...	event
9	GiBleed	anemia	Facial laceration	-inf to -1	4156265	9	9.18	standard	condition...	event
10	GiBleed	anemia	First degree burn	-inf to -1	4296204	8	8.16	standard	condition...	event

```
# i 115 more rows
```

tidy

Note

The `tidy()` function works for any `<summarised_result>` object that we create.

summariseCohortOverlap

When creating multiple cohorts, we might be interested in the overlap of individuals between those cohorts. `summariseCohortOverlap()` does this for you.

```
result <- cdm$my_cohort |>
  summariseCohortOverlap()

glimpse(result)
```

Rows: 72

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name_reference &&& cohort_name_comparator", "cohort_name_reference &&& cohort_na...
$ group_level    <chr> "anemia &&& osteoarthritis", "anemia &&& osteoarthritis", "anemia &&& osteoarthritis", "...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "Only in reference cohort", "In both cohorts", "Only in comparator cohort", "Only in ref...
$ variable_level <chr> "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Subjects", "Sub...
$ estimate_name  <chr> "count", "count", "count", "count", "count", "count", "count", "count", "count", "count", "count"...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", ...
$ estimate_value <chr> "3", "97", "2533", "93", "7", "45", "99", "1", "17", "2533", "97", "3", "2578", "52", "0...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```

tableCohortOverlap

You can also get a table with `tableCohortOverlap()`.

```
tableCohortOverlap(result)
```

Cohort name reference	Cohort name comparator	Estimate name	Variable name		
			Only in reference cohort	In both cohorts	Only in comparator cohort
GiBleed					
anemia	osteoarthritis	N (%)	3 (0.11%)	97 (3.68%)	2,533 (96.20%)
	pneumonia	N (%)	93 (64.14%)	7 (4.83%)	45 (31.03%)
	rheumatoid_arthritis	N (%)	99 (84.62%)	1 (0.85%)	17 (14.53%)
osteoarthritis	pneumonia	N (%)	2,578 (98.02%)	52 (1.98%)	0 (0.00%)
	rheumatoid_arthritis	N (%)	2,612 (99.32%)	18 (0.68%)	0 (0.00%)
pneumonia	rheumatoid_arthritis	N (%)	52 (74.29%)	0 (0.00%)	18 (25.71%)

summariseCohortTiming

When creating multiple cohorts, we might be interested in the timing of entry between cohorts. `summariseCohortTiming()` does this for you.

```
result <- cdm$my_cohort |>
  summariseCohortTiming(restrictToFirstEntry = TRUE)

glimpse(result)
```

Rows: 8,274

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1...
$ cdm_name       <chr> "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", "GiBleed", ...
$ group_name     <chr> "cohort_name_reference &&& cohort_name_comparator", "cohort_name_reference &&& cohort_na...
$ group_level    <chr> "anemia &&& osteoarthritis", "anemia &&& osteoarthritis", "anemia &&& osteoarthritis", "...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
$ variable_name  <chr> "number records", "number subjects", "days_between_cohort_entries", "days_between_cohort...
$ variable_level <chr> NA, NA, NA, NA, NA, NA, NA, NA, "density_001", "density_002", "density_003", "density_004", ...
$ estimate_name  <chr> "count", "count", "min", "q25", "median", "q75", "max", "density_x", "density_x", "densi...
$ estimate_type  <chr> "integer", "integer", "integer", "integer", "integer", "integer", "integer", "integer", "numeric", ...
$ estimate_value <chr> "97", "97", "-17715", "-4915", "2205", "6680", "15530", "-16313.4", "-16251.7980430528",...
$ additional_name <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", ...
```

tableCohortTiming

You can also get a table with `tableCohortTiming()`.

```
tableCohortTiming(result = result,
                  timeScale = "years",
                  uniqueCombinations = FALSE)
```

Cohort name reference	Cohort name comparator	Variable name	Estimate name	Estimate value
GiBleed				
anemia	osteoarthritis	number records	N	97
		number subjects	N	97
		years_between_cohort_entries	Median [Q25 - Q75]	6.04 [-13.46 - 18.29]
			Range	-48.50 to 42.52
	pneumonia	number records	N	7
		number subjects	N	7
		years_between_cohort_entries	Median [Q25 - Q75]	29.90 [15.57 - 38.99]
			Range	1.77 to 53.45
	rheumatoid_arthritis	number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	-17.99 [-17.99 - -17.99]
			Range	-17.99 to -17.99
osteoarthritis	anemia	number records	N	97
		number subjects	N	97
		years_between_cohort_entries	Median [Q25 - Q75]	-6.04 [-18.29 - 13.46]
			Range	-42.52 to 48.50
	pneumonia	number records	N	52

Cohort name reference	Cohort name comparator	Variable name	Estimate name	Estimate value
		number subjects	N	52
		years_between_cohort_entries	Median [Q25 - Q75]	43.49 [38.25 - 50.44]
			Range	16.24 to 61.03
	rheumatoid_arthritis	number records	N	18
		number subjects	N	18
		years_between_cohort_entries	Median [Q25 - Q75]	5.61 [-2.01 - 15.46]
			Range	-18.11 to 34.77
pneumonia	anemia	number records	N	7
		number subjects	N	7
		years_between_cohort_entries	Median [Q25 - Q75]	-29.90 [-38.99 - -15.57]
			Range	-53.45 to -1.77
	osteoarthritis	number records	N	52
		number subjects	N	52
		years_between_cohort_entries	Median [Q25 - Q75]	-43.49 [-50.44 - -38.25]
			Range	-61.03 to -16.24
rheumatoid_arthritis	anemia	number records	N	1
		number subjects	N	1
		years_between_cohort_entries	Median [Q25 - Q75]	17.99 [17.99 - 17.99]
			Range	17.99 to 17.99
	osteoarthritis	number records	N	18
		number subjects	N	18
		years_between_cohort_entries	Median [Q25 - Q75]	-5.61 [-15.46 - 2.01]
			Range	-34.77 to 18.11



Your turn!



Exercise 2 - large scale characteristics for anemia cohort

Can you get the large-scale characteristics from the **drug_exposure** table for the **anemia** cohort table using an anytime-prior time window (excluding index date)? Visualise the result showing the top 5 most frequent codes.

►  [Click to see solution](#)

Top	Estimate value
1	Acetaminophen 325 MG Oral Tablet (1127433) 79 (80.6%)
2	Aspirin 81 MG Oral Tablet (19059056) 71 (72.5%)
3	poliovirus vaccine, inactivated (40213160) 64 (65.3%)
4	Penicillin V Potassium 250 MG Oral Tablet (19133873) 54 (55.1%)
5	Ampicillin 100 MG/ML Injectable Solution (19129655) 9 (9.2%)

CohortCharacteristics

- 👉 Package website
- 👉 CRAN link
- 👉 Manual

